

Lab 12 Activity

PSYC 7804 - Spring 2026

The data comes from an [unpublished master's paper by Carl Hoffstedt](#). They relate the automobile accident rate, in accidents per million vehicle miles to several potential terms. The data include 39 sections of large highways in the state of Minnesota in 1973. Here is a description of the variables:

Variable	Description
rate	1973 accident rate per million vehicle miles
len	length of the Highway1 segment in miles
adt	average daily traffic count in thousands
trks	truck volume as a percent of the total volume
sigs1	(number of signalized interchanges per mile times len + 1)/len, the number of signals per mile of roadway, adjusted to have no zero values.
slim	speed limit in 1973
shld	width in feet of outer shoulder on the roadway
lane	total number of lanes of traffic
acpt	number of access points per mile
itg	number of freeway-type interchanges per mile
lwid	lane width, in feet
htype	An indicator of the type of roadway or the source of funding for the road, either MC, FAI, PA, or MA

Run the following code to name the data you will be using as **dat**:

```
library(carData)
dat <- Highway1
```

1. Run a multiple linear regression with `len`, `slim`, and `acpt` predicting `rate`. What are the significant predictors of `rate` and what is the direction of the relation?
2. Calculate DFBETAS. Is there any observation where the absolute value exceeds 1? If so, which observation/s and for what variables do the absolute values of DFBETAS exceed 1? (the `which()` function may help; run `?which` to bring up the help menu and see what it does)
3. Calculate both DFFITS and Hat Values. For both, find the datapoint with the largest regression diagnostics for both values for both. Is this the same data point that you found in the previous question?
4. Run the same linear regression as before, but remove the outlier that you identified in the previous questions. Do the results change or are they practically the same?
5. Create two added variable plots for `acpt` for:
 - The regression in question 2.
 - The regression in question 4

Do you think that overall the regressions are similar even when removing the largest outlier?